

ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Biometrics, Generative Models

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SELECTED PUBLICATIONS

1. **Allen Tu***, H. Ying*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, ‘[SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
2. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
3. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, 2026.
4. A. Hanson*, **Allen Tu***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 5949–5958.
5. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 21537–21546.

* denotes equal contribution.

EDUCATION

University of Maryland, College Park

Ph.D. in Computer Science

College Park, MD

January 2025 – May 2028

- Advised by Professor Tom Goldstein; collaborating with Matthias Zwicker, Vishal M. Patel, and Rama Chellappa

B.S./M.S. in Computer Science, Minor in Statistics

August 2019 – December 2024

RESEARCH EXPERIENCE

University of Maryland Institute of Advanced Computer Studies

Graduate Research Assistant

College Park, MD

August 2023 – Present

- Introduced pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over $10\times$ fewer primitives while preserving visual fidelity [1, 4, 5]
- Incorporated uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [2]
- Designed scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA) program

Systems & Technology Research

Computer Vision Research Intern

Arlington, VA

May 2025 – January 2026

- Introduced an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [3]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at $1e-3$ FMR using an encoder trained solely on public data [3]
- Developed multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program

Systems & Technology Research

Computer Vision Research Intern

Arlington, VA

May 2024 – August 2024

- Trained a face chip quality estimator via transfer learning, achieving a strong 0.8566 correlation with probe-to-gallery similarity measured in the host SWIN ViT embedding space, outperforming existing methods [3]
- Designed a cluster-and-aggregate network that filters low-importance frame embeddings and fuses the remaining informative ones into a single video template, improving face recognition accuracy by up to 11.54%

Computer Vision Research Co-op

January 2023 – August 2023

- Developed a mixed-voting ensemble that improved open-set search performance by 24.24% for face, 39.18% for body, and 49.10% for fused multimodal recognition
- Trained a self-supervised Barlow Twins model robust to challenges like extreme distance variation and atmospheric interference, improving face recognition accuracy by 52.10% in severe operating conditions

Computer Vision Research Intern

June 2022 – August 2022

- Integrated pose estimation, neural novel-view synthesis, semantic segmentation, and generative inpainting to synthesize realistic cross-subject garment transfers with 12.93 FID and strong biometric identity preservation
- Augmented real training data with garment-transfer images to overcome limited garment diversity and train clothing-invariant whole-body recognition models, improving robustness to cross-garment variation

Undergraduate Researcher

University of Maryland Department of Computer Science

College Park, MD

January 2021 – December 2022

- Analyzed 2.3 million dataset-visualization pairs to identify bias in machine learning visualization recommenders, linking model behavior to user-driven design patterns and proposing data-centric mitigation strategies [6]

RESEARCH PROJECTS

IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA)

University of Maryland Institute of Advanced Computer Studies

August 2023 – Present

Unconstrained 3D reconstruction and novel view synthesis in challenging real-world environments.

- Efficient rendering, compression, and training methods for 3D and 4D Gaussian Splatting (3DGS/4DGS) [1, 4, 5]
- Multi-view consistent super-resolution for training 3DGS with low-resolution imagery [2]
- Image, video, and multi-view diffusion model priors for sparse-view 3D reconstruction
- Uncertainty quantification algorithms for 3DGS and Neural Radiance Fields (NeRF) [4]

IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR)

Systems & Technology Research

June 2022 – January 2026

Multimodal, opportunistic fusion of incomplete face, body, and gait information in severe operational conditions.

- Transfer Learning for Face Image Recognizability Assessment (2025) [3]
- Learned Frame Feature Aggregation for Face Recognition with Low-Quality Video (2024)
- Operating Condition-Invariant Barlow Twins and Multimodal Ensembling (2023)
- Style-Based Appearance Flow for Clothing-Robust Body Representation Learning (2022)

SERVICE

SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models

Workshop Co-organizer

CVPR 2026

- Co-organized the inaugural CVPR workshop on security, privacy, and robustness in 3D generative vision, coordinating submissions, reviewer recruitment, the peer review process, and industry sponsorship

Professional Memberships: IEEE Member; Computer Vision Foundation (CVF); IEEE Biometrics Council

TECHNICAL SKILLS

Languages: Python, CUDA, C/C++, Shell, Java, SQL, MATLAB

Libraries: PyTorch, TensorFlow, NumPy, SciPy, OpenCV, Matplotlib, scikit-learn, pandas, Unity

Research: 3D/4D Gaussian Splatting, Neural Radiance Fields (NeRF), Differentiable Rendering, Diffusion Models, Vision Transformers (ViT/Swin), Multimodal Encoders, Uncertainty Quantification, Super-Resolution, GPU Programming, Structure-from-Motion (SfM), Mixture-of-Experts (MoE), Large Language Models (LLMs), Virtual Reality (VR)

Selected Graduate Coursework: Advanced Computer Graphics, Advanced Numerical Optimization, Advances in XR, Computational Geometry, Foundations of Deep Learning, Physically-based Modeling & Simulation & Animation, Natural Language Processing, Interactive Technologies in Human-Computer Interaction

Selected Undergraduate Coursework: Advanced Data Structures, Applications of Linear Algebra, Applied Probability and Statistics, Capstone in Machine Learning, Computer Vision, Database Design, Design and Analysis of Algorithms, Introduction to Data Science, Introduction to Machine Learning, Introduction to Parallel Computing, Introduction to Probability Theory, Introduction to Statistical Computation, Programming Language Technologies and Paradigms

TEACHING EXPERIENCE

Peer Research Mentor (Capital One Machine Learning)

College Park, MD

First Year Innovation and Research Experience (FIRE)

January 2021 – December 2022

- Mentored 80 undergraduate students across two cohorts in a selective multi-semester machine learning research program, guiding students from foundational training to independent research project execution
- Designed and delivered a project-driven curriculum spanning data processing, model development, training, evaluation, and deployment using modern deep learning frameworks and recent research advances
- Advised student teams end-to-end on collaborative research projects, leading to presentations in areas including text-to-audio generation, image super-resolution, and face attribute recognition